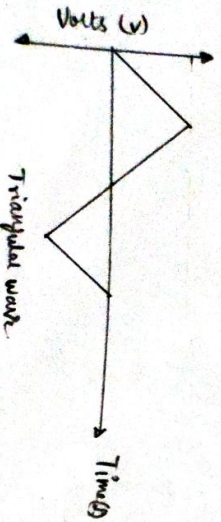
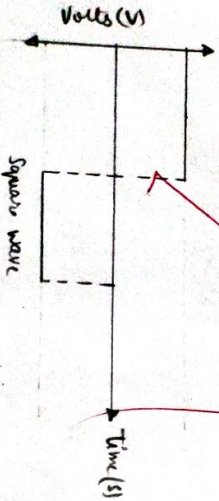
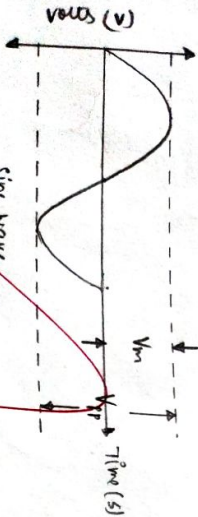


Table 1

Sl. No	Waveform	Amplitude	Time period	Frequency or signal generator (KHz)	Calculated Frequency $F = \frac{1}{T}$ (KHz)
1.	Sine	$3.6 \times 1 = 3.6 \text{ mV}$	$2 \times 2 = 4 \mu\text{s}$	30	$\frac{1}{4} = 0.25 = 250 \text{ KHz}$
		$3.6 \times 1 = 3.6 \text{ mV}$	$1.9 \times 1 = 1.9 \mu\text{s}$	600	$\frac{1}{1.9} = 0.526 = 526 \text{ KHz}$
2.	Square	$3.6 \times 1 = 3.6 \text{ mV}$	$2.6 \times 1 = 2.6 \mu\text{s}$	400	$\frac{1}{2.6} = 0.38 = 380 \text{ KHz}$
		$3.6 \times 1 = 3.6 \text{ mV}$	$2.6 \times 0.5 = 1.3 \mu\text{s}$	800	$\frac{1}{1.3} = 0.76 = 760 \text{ KHz}$
		$3.6 \times 1 = 3.6 \text{ mV}$	$2 \times 0.5 = 1 \mu\text{s}$	1000	$\frac{1}{1} = 1 = 1000 \text{ KHz}$
3.	Triangular	$3.6 \times 1 = 3.6 \text{ mV}$	$2.6 \times 0.2 = 0.52 \mu\text{s}$	8000	$\frac{1}{0.52} = 1.9 = 1900 \text{ KHz}$

Master graph:



1

STUDY OF CRO

17-6-22

Aim:

To study the operation and the usage of Cathode Ray Oscilloscope and to measure the AC voltage, amplitude and frequency. Equipment and components required:

Sl. NO	Name of the component	Range	No. of required.
1.	Dual channel Cathode Ray Oscilloscope	0-50 MHz	1
2.	Function generator	0-3 MHz	1
3.	BNC Cables and probes	-	3.

Formula used:

$$\text{Peak-to-peak} = \frac{\text{Volt}}{\text{Division}} \times \text{No. of divisions (VOLT)}.$$

Theory:

Cathode ray oscilloscope is a common laboratory instrument that provides accurate time and amplitude measurements to voltage signals over a wide range of frequencies. Its reliability, stability and ease of operation make it an suitable a general purpose laboratory instrument.

The CRO provides a visual representation of any waveform applied to input terminal. The oscilloscope consists of following subsystem.

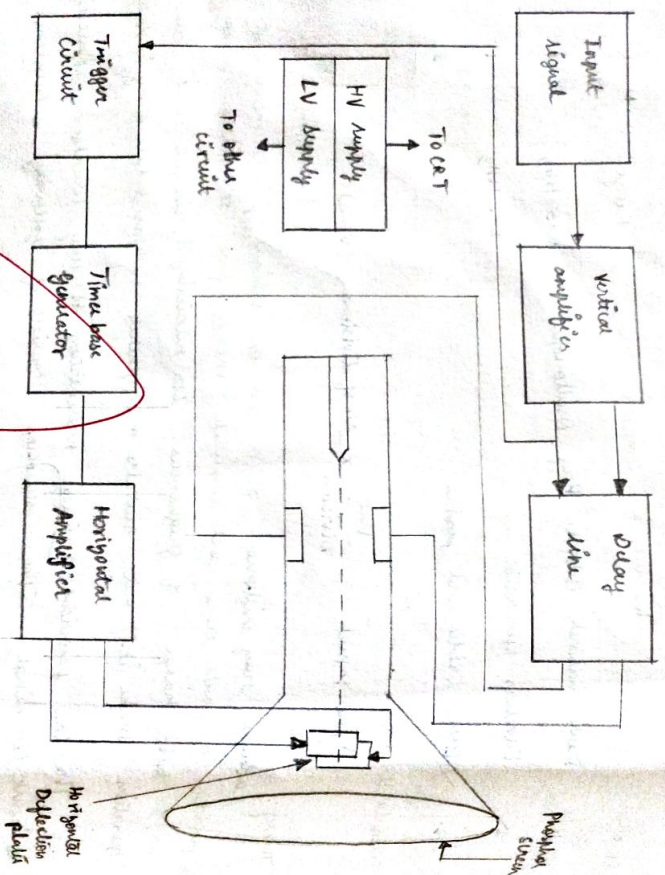
- Cathode ray tube (CRT)
- Vertical amplifier
- Horizontal amplifier
- Sweep generator
- Trigger circuit
- Modulation power supply.

It can be employed to measure quantities such as peak voltage, frequency, phase difference, pulse width, time rise and time fall.

Cathode ray tube (CRT):

- It is the heart of CRO providing visual display of input wave form. It contains four basic parts.
- An electron gun providing a stream of electrons
- Focusing and astigmatism elements to produce well defined beam of electrons
- Horizontal and vertical deflection plates to control path of electron beam.

Block diagram



- An evacuated glass envelope with a phosphorescent screen which glows visibly when struck by electron beam.
A Cathode ray containing oxide is being is heated directly by electron beam.

A cathode containing oxide coating is heated indirectly by a filament resulting in release of electron cathode and thus control the no. of electrons directed to screen. Once electron passes the control grid, they are focused into a tight beam and accelerated to higher velocity and by focusing and accelerating anodes. The high velocity and well-defined electron beam then passed to through two sets of deflection plates.

The first set of plate is oriented to deflect the electron beam vertically. The anode of vertical deflection is determined by the voltage polarity applied to the deflection plate. The electron beam is also deflected by high voltage applied to horizontal deflection plates. The rule of positivity to deflecting voltage can be expressed in two ways that are deflection factor and deflection sensitivity.

Control grid:

- Regulates the number of electrons that reach the anode and hence the brightness of the spot on the screen.

Focusing anode:

- Ensures that electrons leaving the cathode in slightly different directions are focused down to a narrow beam and all arrive at same spot on the screen.

Electron gun:

- The assembly of the cathode, focusing grid focused and accelerated anode (positive electrode) is called electron gun. Its purpose is to generate the electron beam and control the intensity and focus. Between two electron gun and the phosphor screen, there are two vertical deflections of beam. These plates are thus supposed to as the horizontal and vertical deflection plates. The combination of these deflections allow the beam to reach any portion of the phosphor screen. Whenever the electron hits the screen, phosphor is excited and light is emitted. In general, various microscope amplifier circuits are needed to increase the input signals to the voltage.



Horizontal amplifier:

- Develop a voltage at horizontal deflection signals at its input prior to the signal being applied to the horizontal deflection plates.

Sawtooth generator:

- Develop a voltage at horizontal deflection plate that increase linearly and with time.
- The cathode ray is a beam of electrons which are emitted by heated cathode (negative electrode) and accelerated towards anode.

Operation:

The four main parts of cathode ray or CRT are designed to create and direct electron beam to a screen to form an image. The microscope relies to a circuit that directly controls the vertical deflection plates while the horizontal plates have linearly increasing voltage to form the circuit voltage over time. The scanning spot, the area gives electrons in the cathode enough energy to escape the electrons are attracted to the accelerating anode and pulled through a control grid that regulates the number of electrons in beam. The vertical and horizontal deflection plates create electric field that bend of electrons. The electrons finally hit fluorescent screen, which converts the energy from the electron beam and emits in the form of light to display an image at end of glass tube.

Procedure:

- Switch on CEO. Turn on the DC supply. Check if a horizontal (line) trace appears after CEO warms up. Set the focus centrally in the position on the screen.

Measurement of frequency:

- Connect a signal generator output to the one vertical deflection of CEO.
- Set the function generator in the sinusoidal mode and adjust the amplitude of signal just about to fit the screen.
- At the generator dial at any particular frequency and note the dial until you have a full complete cycle across CEO from its horizontal direction.
- Measure the period T of the signal to do so measure horizontal distance between two successive peaks and multiply by reading of 'time/div' button which scale time axis record your data.
- This gives time period T of AC signal, its frequency is $f = 1/T$ (Hz).



Calculations:

a) Measurement of sine wave:

Amplitude = $3.6 \text{ div} \times 1 = 3.6 \text{ V}$; Amplitude = 3.6 V

Time period for } = $2 \text{ div} \times 2 \text{ V/div}$
 wave form
 $T = 4 \mu\text{s}$

Frequency, $f = \frac{1}{T} = \frac{1}{4} = 0.25 \text{ MHz}$

$f = 250 \text{ KHz}$

b) Measurement of square wave:

Amplitude, $A = 3.6 \text{ div} \times 1 \text{ V/div}$

$A = 3.6 \text{ V}$

Time period for } $T = 2.6 \text{ div} \times 1 \text{ V/div}$
 wave form
 $T = 2.6 \mu\text{s}$

Frequency, $f = \frac{1}{T} = \frac{1}{2.6} = 0.38 \text{ MHz}$

$f = 380 \text{ KHz}$

c) Measurement of Triangular wave:

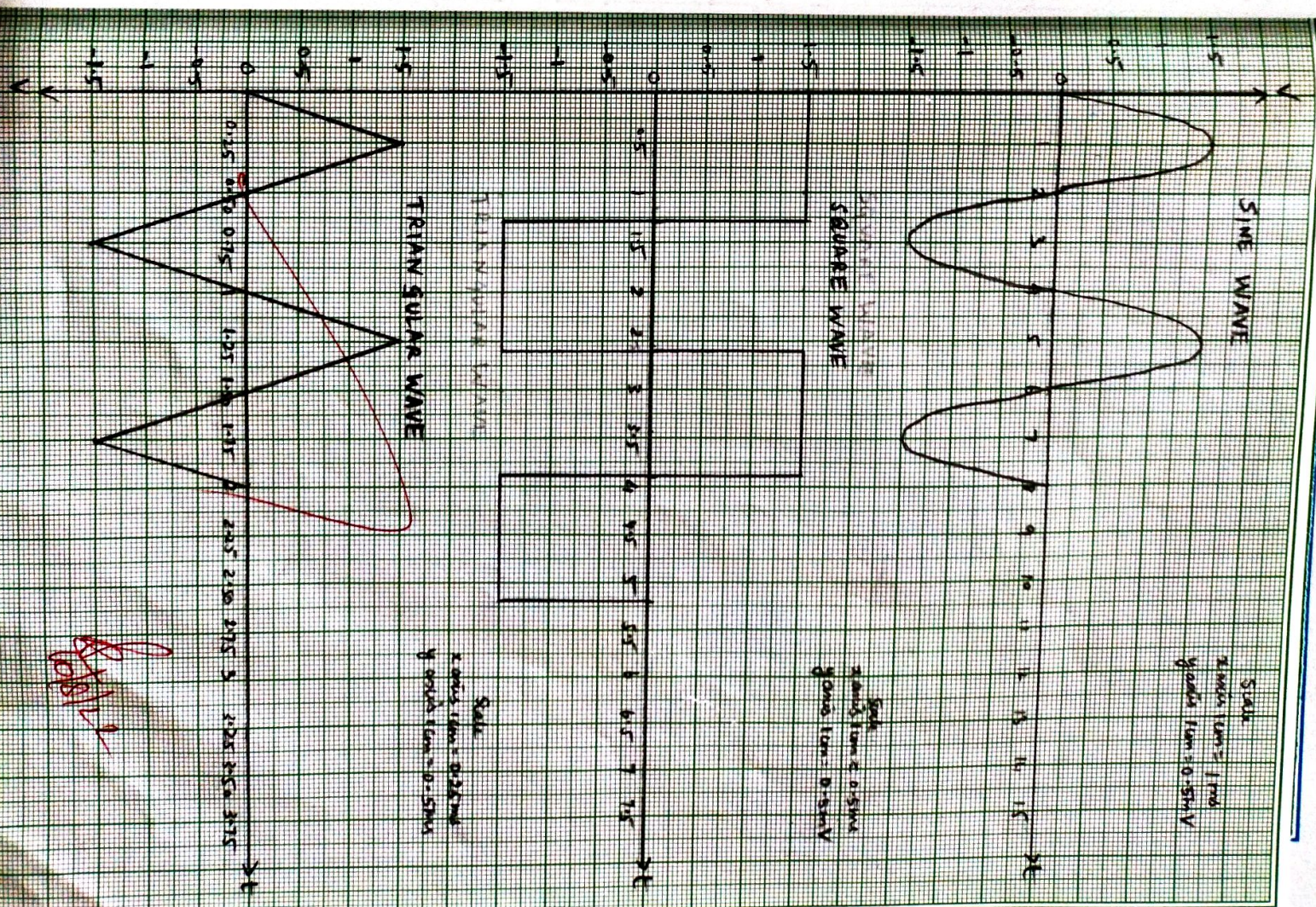
Amplitude, $A = 3.6 \text{ div} \times 1 \text{ V/div}$

$A = 3.6 \text{ V}$

Time period for } $T = 2 \text{ div} \times 0.5 \text{ V/div}$
 wave form
 $T = 1 \mu\text{s}$

Frequency, $f = \frac{1}{T} = \frac{1}{1} = 1 \text{ MHz}$

$f = 1000 \text{ KHz}$



Measurement of voltage :

- Use the voltage division selector to convert voltage readings on the multi meter into actual voltages.
- In measuring the voltage, always measure the value from across circuit from its peak. This "peak voltage" is half the peak-to-peak voltage which is the full height of trace on the CRO screen.
- Repeat the above procedure of frequency and voltage measurement for sinusoidal signal, triangular and square wave also.

Result :

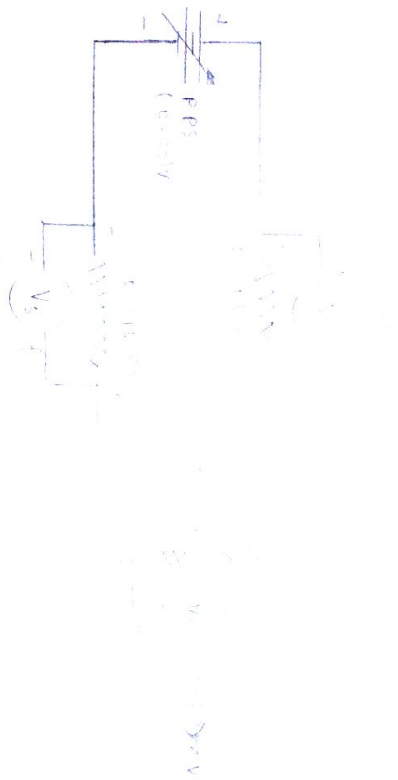
Thus amplitude and frequency of sine, square and triangular wave form was studied with help of cathode ray oscilloscope (CRO).

20/11/22

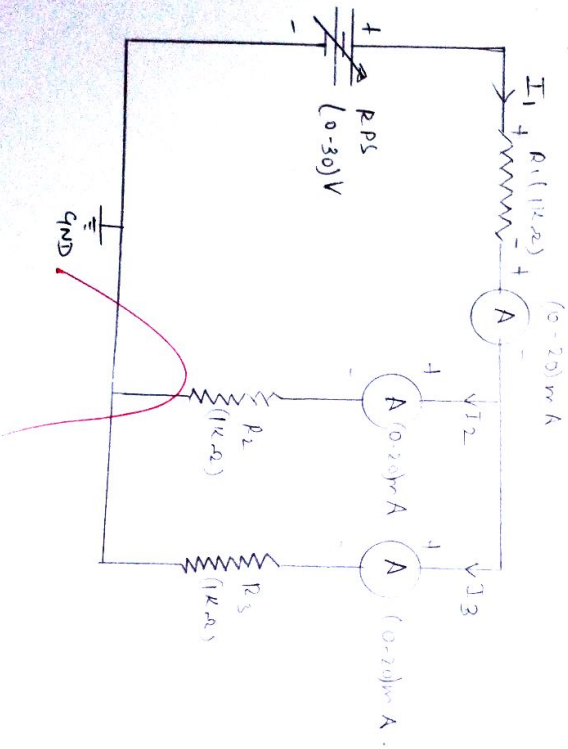


Circuit Diagram

Kirchoff's Voltage Law:



Kirchoff's Current Law:



2

VERIFICATION OF KIRCHOFF'S LAWS

8-1-22

Aim:

To verify Kirchhoff's voltage law and current law using simple circuit. Equipment and components required:

Sl. No	Name of the Component	Range	Quantity
1.	Regulated power supply	0-30 V	1 No.
2.	DC voltmeter	0-30 V	2 Nos.
3.	DC ammeter	0-50 mA	2 Nos.
4.	DC Ammeter	0-100 mA	1 No.
5.	Resistors	1K Ω	3 Nos.
6.	Bread Board	-	1 No.
7.	Connecting wire	-	As required.

Law and Formula Used:

KVL: The algebraic sum of current at any junction of network is zero. In other words, algebraic sum of current leaving is equal to algebraic sum of current entering a junction.

In given circuit, the current at point B is given by:

$$I_1 - I_2 - I_3 = 0 \quad (\text{or}) \quad I_1 = I_2 + I_3$$

KVL: In any closed circuit, the algebraic sum of potential drop ($I \cdot R$) is equal to impressed emf.

In given circuit, the emf is given by:

$$E = V_1 + V_2 \quad (\text{or}) \quad E = V_1 + V_3$$

Precautions:

- Power switch of EPS should be in OFF position at time of starting.
- Connection must be tight and tight.
- True should not be any error in meters.
- Polarity of ammeter and voltmeter should be connected correctly in order to avoid damage to it.
- Loose and tight adjustment knobs in EPS should be kept at minimum position at time of starting.

Procedure:

- Observe the circuit diagram.
- Decide values of R_1 , R_2 and R_3 resistors.
- Also decide the input voltage for which you are going to verify the laws by theoretically and practically.



For KYL:

- Calculations are made to put the circuit diagram
- observing the measurements, switch on the RPS.
- Set the RPS in volt position and check across if any.
- At the voltage at 'zero' in the display with the help of coarse and fine adjustments knobs in RPS.
- The input voltage is slowly adjusted and noted in the readings of all meters as P_1 (volmeter), P_2 (wattmeter) and P_3 (ammeter).
- Tabulate the readings and verify whether input voltage is equal to sum of voltage drop in the circuit that is $E = V_1 + V_2 + V_3$.
- Now the RPS is brought to minimum position and switch off the unit.

For KEL:

- Connections are made as per the circuit diagram
- Observe the measurements. Switch on the RPS
- At the RES voltage and check if any across if any
- At the voltage at 'zero' in the display with help of coarse and fine adjustments knobs in RPS.
- The input voltage is adjusted slowly and note down the E of all meter such that $E = I_1 (R_1 + R_2 + R_3)$ (ammeter) and P_3 (ammeter).
- Tabulate the readings and verify whether the current is at point B.
- is equal to zero, that is $I_1 = I_2 + I_3$.
- Now the RPS is brought to minimum position and switch off the unit.

Theory:

Network element can be either active or passive type of electrical circuit (as network contain one of them or a combination of both).

Two laws of Kirchhoff's laws are

- Kirchhoff's Current Law
- Kirchhoff's Voltage Law

Kirchhoff's Current Law (KCL):

- It states that algebraic sum of currents leaving or entering a node is zero.
- A node is a point at which two or more circuit elements connected to it.
- If only two circuit elements are connected to a node, then it is said to be a simple node. If three or more circuit elements are connected to a node, it is called as multiple node.

Tabulation:

1. Kirchhoff's Voltage Law (KVL):

Sl.No	Applied Voltage (V)	Voltage across R_1 (V)		Voltage across R_2 (V)		Voltage across R_3 (V)		Total voltage (V)	
		Theoretical	Practical	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1.	3V	1	1.04	1	1.01	1	1.02	3	3.07
2.	6V	2	2.04	2	1.97	2	2.01	6	6.02
3.	9V	3	3.05	3	2.98	3	2.98	9	9.01

2. Kirchhoff's Current Law (KCL):

Sl.No	Applied Voltage (V)	Current through R_1 (I ₁)		Current through R_2 (I ₂)		Current through R_3 (I ₃)		Total Current (I)	
		Theoretical	Practical	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
1.	3.17	3.17	1.59	1.59	1.55	1.58	3.1	3.17	
2.	6.06	1.05	3.03	3.02	3.03	3.02	6.06	6.04	
3.	9	9	4.5	4.5	4.5	4.49	9	9.02	

I_m is the branch current leaving the node m is the node branches that are connected to a node.

- The above statement KCL also is expressed as algebraic sum of currents entering a node is equal to algebraic sum of currents leaving a node.

Let us verify through the statement through the branch current I_1, I_2 and I_3 entering a node P. So consider a negative sign for the currents.

- In the above figure the branch currents current I_4 and I_5 are leaving from node P. So consider positive sign for the currents.

$$\rightarrow -I_1 - I_2 - I_3 + I_4 + I_5 = 0 \rightarrow I_1 + I_2 + I_3 = I_4 + I_5$$

The left hand side represents the sum of entering elements whereas right hand side represents the sum of leaving currents.

- In this circuit the positive sign is considered. When the current is leaving node and entering node is negative and positive sign respectively. In both cases the result will be the same.

Kirchhoff's Voltage Law:

- It states that algebraic sum of voltages around a loop or mesh is equal to zero. A loop is a path that terminates at the same node when it is started from, if constant, a mesh is a loop that doesn't contain any other loop inside it.

- The algebraic sum of voltages around a loop is equal to algebraic sum of voltage drops that are present in a loop. Let us verify through the statements by

$$\rightarrow -V_1 - V_2 - V_3 + V_4 + V_5 \rightarrow V_1 + V_2 + V_3 = V_4 + V_5$$

- In above equation the left hand side represents voltage drops and right hand side represents the voltage increase. Here we consider only single voltage source that produces voltage.

- In this circuit we consider the sign of each element voltages as the voltage of the source terminal that is present while travelling around the loop similarly you consider the sign of each voltage as the polarity of first terminal that is present while travelling around the loop in both ways.



Theoretical

At 3V,

$$E = V_1 + V_2 + V_3 = 1 + 1 + 1 = 3$$

$$E = 3 \text{ mV}$$

At 6V,

$$E = 6 \text{ mV}$$

At 9 V,

$$E = 9 \text{ mV}$$

Practical

At 3V,

$$E = V_1 + V_2 + V_3 = 1.04 + 1.01 + 1.02$$

$$E = 3.07 \text{ mV}$$

At 6V,

$$E = 6.02 \text{ mV}$$

At 9V,

$$E = 9.01 \text{ mV}$$

By KVL:

$$R_T = R_1 + (R_2 \parallel R_3)$$

$$= 1 + \frac{1 \times 1}{1+1}$$

$$R_T = 1.5$$

→ For KCL, the current remains constant,

→ By Ohm's Law

$$V = IR$$

$$I = \frac{V}{R} = 0.66 \text{ A}$$

$$I = 0.66 \text{ A}$$

Theoretical

To find R_T

$$R_T = \frac{V}{I}$$

$$= \frac{1}{0.66} = 1.5$$

$$R_T = 1.5 \Omega$$

Then,

$$I_2 = I$$

Theoretical

$$V = 3 \text{ V}, I = 0.66 \text{ A}$$

To find R_2 ,

$$R_2 = \frac{V_2}{I} = \frac{3}{0.66}$$

$$R_2 = 4.5 \Omega$$

To find R_3 , theoretical

$$R_3 = \frac{V_3}{I} = \frac{3}{0.66}$$

$$R_3 = 4.5 \Omega$$

Practical

$$R_1 = R_2 + R_3$$

$$= 1.69 + 1.58$$

$$R_1 = 3.17 \Omega$$

$$I_1 = \frac{V}{R_1}$$

$$= \frac{1}{1.69}$$

$$I_2 = \frac{1}{1.58}$$

$$= 1$$

$$R = \frac{V}{I}$$

$$= \frac{1}{0.66}$$

$$I_1 = I_2 + I_3$$

Practical

$$R_1 = R_2 + R_3$$

$$R_2 = 4.55 + 4.49$$

$$R_2 = 9.02$$

Practical

$$R_3 = R_1 + R_2$$

$$= 4.5 + 4.5$$

$$R_3 = 9 \Omega$$

$$V = IR$$

$$I = \frac{V}{R} = \frac{3 \text{ V}}{1 + (1/1)} = 1 \text{ mA}$$

$$V = V_1 + V_2 + V_3$$

$$V_1 = I \cdot R_1$$

$$V_2 = I \cdot R_2$$

$$V_3 = I \cdot R_3$$

$$V_1 = 1 \times 1$$

$$V_2 = 1 \times 1$$

$$V_3 = 1 \times 1$$

$$V = 6 \text{ V}$$

$$I = \frac{V}{R} = \frac{6}{3}$$

$$= 2$$

$$V_1 = 2 \times 1$$

$$V_2 = 2 \times 1$$

$$V_3 = 2 \times 1$$

$$= 2$$

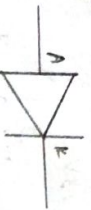
Result:

Thus the theoretical current law (KCL) and voltage law (KVL) of given circuit is verified.

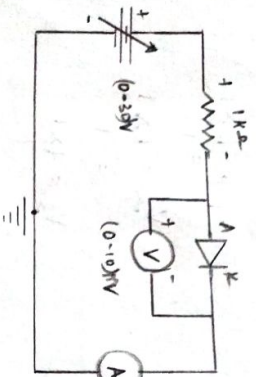
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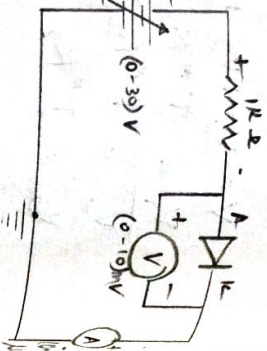
Circuit diagram for PN Junction diode:
Symbol for PN Junction diode:



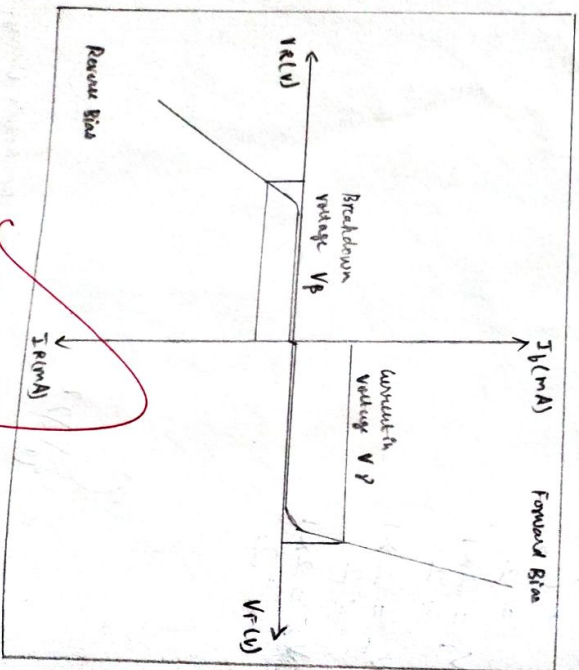
Forward Bias:



Reverse Bias:



Model graph:



3

V-I CHARACTERISTICS OF PN JUNCTION DIODE

29-1-22

Aim:

To determine the forward and reverse bias V-I characteristics of PN Junction diode.

Equipments and components required:

Sl. No	Name of the component	Specifications	Range	Quantity
1.	PN Junction diode	1N4007	-	1
2.	Resistor	-	100 Ω	1
3.	Regulated power supply	-	(0-30) V	1
4.	Voltmeter	-	(0-1) V, (0-10) V	2
5.	Ammeter	-	(0-500) mA	2
6.	Bread Board	-	-	2
7.	Connecting wires	-	-	-

Formula used:

Forward static resistance = $\frac{V_F}{I_F}$; Forward dynamic resistance = $\frac{dV_F}{dI_F}$
Reverse static resistance = $\frac{V_R}{I_R}$; Reverse dynamic resistance = $\frac{dV_R}{dI_R}$

Theory:

A single junction and two terminal device. A PN junction diode is formed by a layer of P-type and N-type semiconductor. When a PN junction is formed, the free electrons in N-region diffuse across the junction and combine with the holes in P-region and a depletion layer is developed. The depletion layer consists of ions in which acts as barrier beyond a certain limit. The diagram of potential across depletion layer called the potential barrier. At 25°C, the barrier potential approximately equals 0.7V for silicon and 0.3V for germanium diode.

Ge diode has smaller cut-in voltage and when compared to Si diode. The reverse saturation current in Ge diode and longer in magnitude when compared to Si diode.

PN Junction connected to an external voltage source is called a biased PN junction. A PN junction diode conducts only in one direction. The V-I characteristics of the diode are curve between across the diode and current through the diode.

When the circuit voltage is zero. circuit is open and potential barrier doesn't allow the current through to flow across it, hence the circuit current is zero.



Tabulation :

Forward Bias :

M.No	Voltage across diode (V)	Current through diode (mA)
1.	0.1	0
2.	0.2	0
3.	0.3	0
4.	0.4	0.02
5.	0.5	0.13
6.	0.5	0.34
7.	0.55	0.88
8.	0.6	309
9.	0.65	5.66
10.	0.7	16.57

Reverse Bias :

M.No.	Voltage across diode (V)	Current through diode (mA)
1.	1	0.1
2.	2	0.2
3.	3	0.3
4.	4	0.4
5.	5	0.5
6.	6	0.6
7.	7	0.7
8.	8	0.8
9.	9	0.9
10.	10	1

When P-type (anode) is connected to positive terminal and N-type is connected to negative terminal to the supply voltage, it is known as forward bias. The barrier potential is reduced when diode is forward biased, thus the barrier potential is completely eliminated and current starts flowing through the diode in the circuit. The diode is said to be in ON state. The current increases with increase in forward voltage.

When N-type (cathode) is connected to positive terminal and P-type is connected to negative terminal to the supply voltage, it is known as reverse bias. The barrier potential is increased. Therefore the junction resistance becomes very high and a small current (reverse saturation current) flows across the circuit. The diode is said to be in OFF state. The current decreases due to minority carriers.

Procedure :

Forward Bias :

- Connections are made as per circuit diagram.
- For forward bias, the PPS terminal is connected to the anode of diode and PPS is connected to cathode of diode.
- Switch on the power supply and increase input voltage (supply voltage) in steps.
- Note down the corresponding current flowing through the diode and voltage across the diode for each and every step of input voltage.
- Graph & plotted between voltage and current.
- The readings of voltage and current are tabulated.

Reverse Bias :

- Connections are made as per the circuit diagram.
- For reverse bias, PPS terminal is connected to the cathode of diode and PPS is connected to anode of diode.
- Switch on the power supply and increase input voltage (supply voltage) in steps.
- Note down the corresponding current flowing through the diode and voltage across the diode for each and every step of input voltage.
- The readings of voltage and current are tabulated.
- Graph & plotted between voltage and current.



$$\frac{5.65}{19.5} = \frac{V}{1}$$

$$\frac{5.65}{19.5} = \frac{V}{1}$$

$$R = 0.1142$$

Dynamic Forward Kinkness = $\frac{\Delta V}{\Delta I} = \frac{0.65 - 0.55}{5.66 - 0.89}$

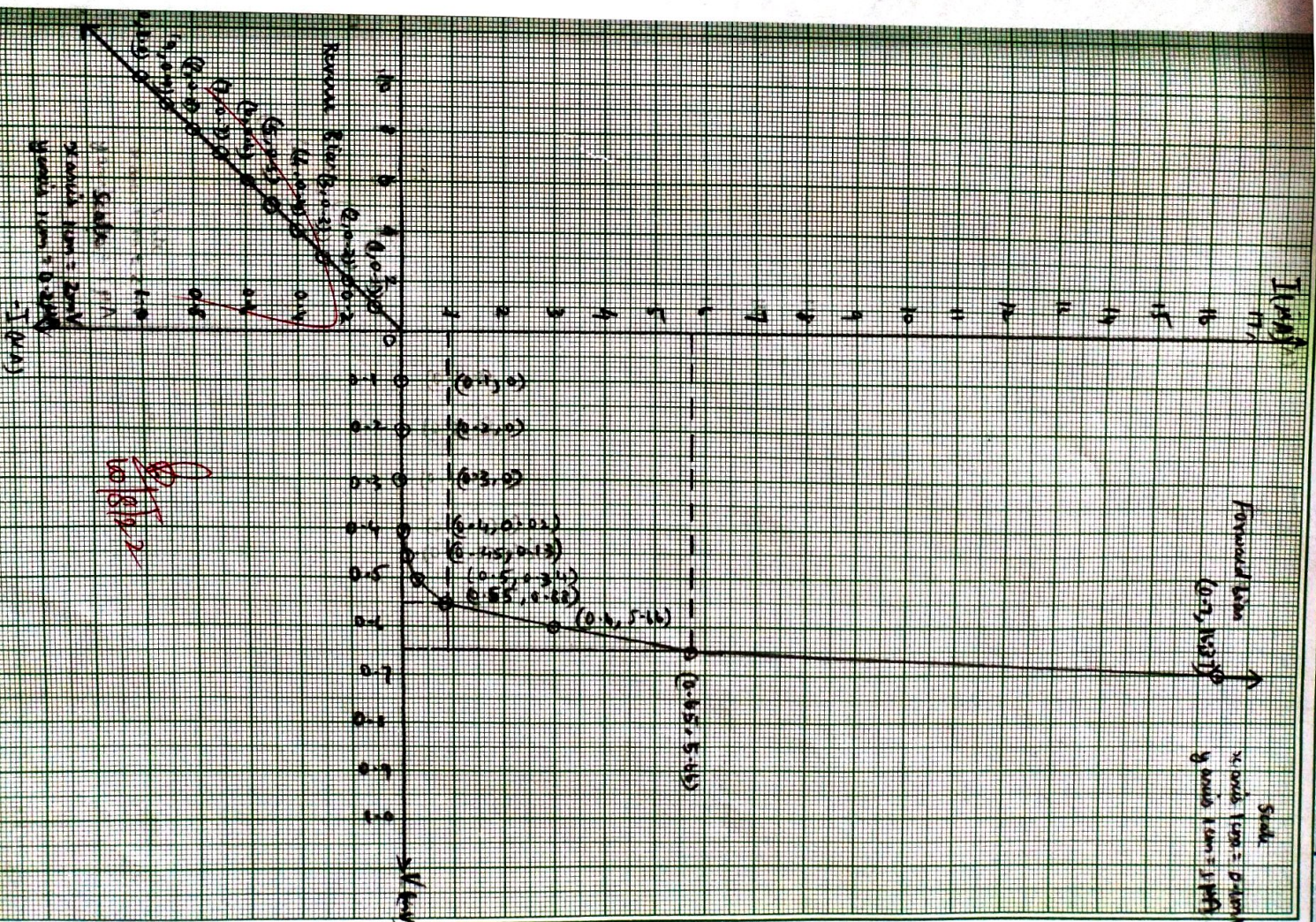
$$R_D = 0.02 \text{ } \Omega$$

static pressure resistance = $\frac{V}{I} = \frac{5}{0.5}$

$$R_F = 10\Omega$$

$$\text{Dynamic Pressure} = \frac{\Delta V}{\Delta t} = \frac{5-3}{0.5-0.3}$$

~~$R_D = 10\Omega$~~



Result: True DC V-I characteristics of Si-PN Diode is plotted and verified.

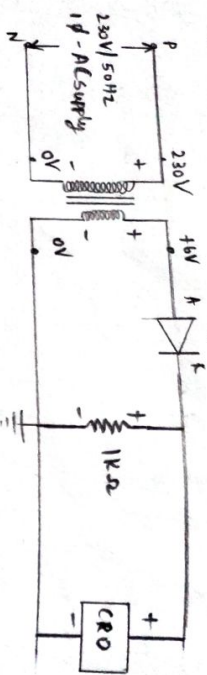
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Thermal, Static Forward Resistance	= 0.114 Ω
Dynamic Forward Resistance	= 0.02 Ω
AC static Reverse Resistance	= 10 Ω
Dynamic Reverse Resistance	= 10 Ω

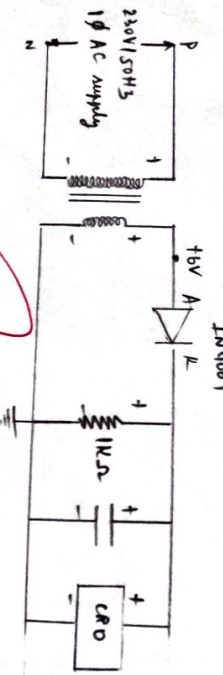


Result diagram :

Half wave rectifier without filter :



Half wave rectifier with filter :



4

APPLICATION OF PN JUNCTION DIODE AS
HALF WAVE RECTIFIER

S-8-22

Aim :

To design a half wave rectifier circuit using PN junction diode and measure ripple factor with and without capacitor filter.

Equipment and components required :

Sl.No	Apparatus Required	Range	Quantity
1.	1 φ transformer	230V (6-0-6)V, 150mA, 50Hz	1 No.
2.	PN junction diode	-	1 No.
3.	Resistor	1KΩ	1 No.
4.	Capacitor	100μF	1 No.
5.	DC Voltmeter	(0-30)V	1 No.
6.	Cathode ray oscilloscope (CRO)	(0-30) MHz	1 No.
7.	Bread board	-	1 No.
8.	Power chords	-	1 No.
9.	Probe	-	1 No.
10.	Connecting wires	-	As required.

Formula Used :

$$\text{Ripple factor} = \frac{V_{ac}}{V_{dc}} \quad \text{where } V_{ac} = \text{rms voltage of AC components of average output voltage,}$$

$V_{dc} = \text{voltage of DC component}$

Half wave rectifier without filter :

$$V_{rms} = \frac{V_m}{\sqrt{2}} \text{ and } V_{dc} = \frac{V_m}{\pi}$$

$$(r), \text{ Ripple factor} = \frac{\sqrt{V_{rms}^2 - V_{dc}^2}}{V_{dc}} = 1$$

Theoretical method :

$$r = \frac{V_{ac}}{V_{dc}}$$

Half wave rectifier with filter :

$$V_{dc} = V_m - \frac{V_r}{2} \text{ and } V_{rms} = \frac{V_r}{2\sqrt{2}}$$

$$(r), \text{ Ripple factor} = \frac{V_{rms}}{V_{dc}}$$

Theoretical method :

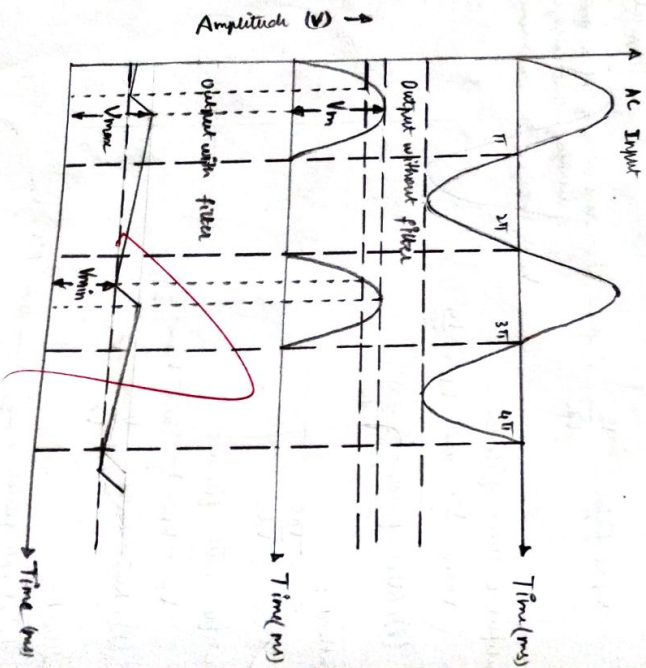
$$(r), \text{ Ripple factor} = \frac{1}{2\sqrt{3}(4CRf)} \text{ and } PIV = V_m$$



Tabulation: Half wave rectifier with PN junction diode:

Input		without filter		Output with filter	
Amplitude (V)	Time period (ms)	Amplitude (V)	Time period (ms)	Amplitude (V)	Time period (ms)
3.875 = 19mV	475 = 20ms	1.675 = 8mV	475 = 20ms	$V_{max} = 0.875$ $V_{min} = 0.255$ $= 1mV$	$V_{max} = 475$ $V_{min} = 475$ $= 20mV$

Model graph (Half wave rectifier):



Theory:

Half wave rectifier:

A rectifier is a circuit which uses one or more diodes to convert AC voltage to pulsated DC voltage. In this rectifier circuit step-down transformer is used at the input side. The transformer couples an AC input voltage to the rectifier circuit. A diode is connected between secondary of the transformer and the load resistor.

Operation of half wave rectifier:

During positive half cycle of AC input, the diode is forward biased and conducts for all voltages greater than the offset voltage of semiconductor material used. The voltage produced across load resistor has same shape of the positive input half cycle of AC input voltage.

During negative half cycle of AC input, the diode is reverse biased and doesn't conduct for all voltages greater than the offset voltage of semiconductor. The net result is that positive half cycle of the input voltage drops across load resistor.

The net result is that only positive half cycle of the input voltage appears at output.

Precautions:

- Power switch should be kept at OFF position at the time of starting.
- Connections must be neat and tight.
- There should not be any wires in multimeter.
- Relating to voltmeter should be connected correctly in order to avoid the damage to it.
- Channel 1 and Channel 2 of CRO should be kept at ground position at time of starting.

Procedure:

- Connection are made as per circuit diagram.
- The resistor will act as load to output of PN junction diode.
- Connect the CRO channel 1 probe to output of PN junction diode. That is parallel to resistor.
- Measure the pre-amplified output on power supply.
- Turn on CRO and observe a horizontal line on the screen. Ensure all settings are unchanged. You are looking at channel 1.
- Set CRO at AC position and now half the sine wave will appear as in model graph.
- Note down the reading of V_m from DC voltmeter and V_{ac} from CRO.



Calculations:

Half wave rectifier without filter:

$$V_{rms} = \frac{V_m}{2} = \frac{230}{2} = 115 \text{ V} ; V_{dc} = \frac{V_m}{\pi} = \frac{230}{\pi} = 73.21 \text{ V}$$

$$\text{Ripple factor } (\gamma) = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{115}{73.21}\right)^2 - 1}$$

$$\gamma = 1.211$$

Half wave rectifier with filter:

$$r = \frac{V_{ac}}{V_{dc}} = \frac{230}{6} = 38.333$$

$$V_{dc} = V_m - \frac{V_L}{2} = 230 - \frac{38.333}{2} = 210.833$$

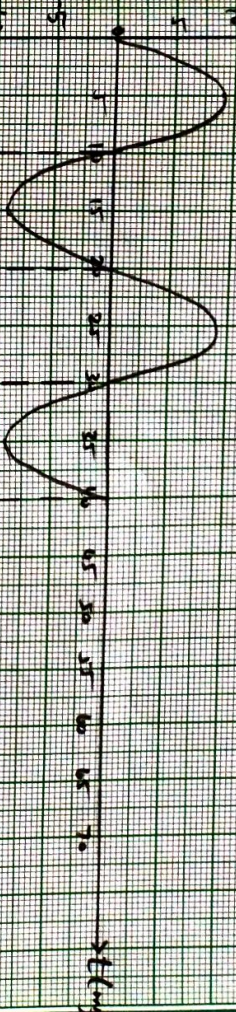
$$V_{rms} = \frac{V_L}{2.43} = 11.065$$

$$\text{Ripple factor } (\gamma) = \frac{V_{rms}}{V_{dc}} = \frac{11.065}{210.833}$$

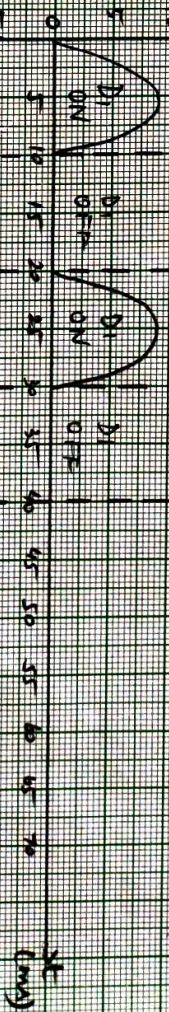
$$\gamma = 0.052$$

Scale
Y axis 1 cm = 5mV
X axis 1 cm = 5ms

AC Input



DC Output (without filter)



DC Output (with filter)



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- Trace the wave from the screen and plot the graph.
- Put the channel 1 in ground position and switch off the power supply.
- Now connect the filter that is capacitor across the resistor as shown in figure.
- Switch on the power supply.
- At the CRO channel 1 at AC position and now half line wave with pulse setting it will appear as shown in next graph.
- Note down the reading from V_{AC} - voltmeter and V_{RC} - CRO.
- Trace wave from screen and plot graph.
- Practical ripple factor can be calculated from the graph.
- Put the CRO channel 1 in ground position and switch off the power supply.

Result :

Thus the application of a PN junction diode as half wave rectifier is analysed and calculate the theoretically and practically value of ripple factor.

Half wave rectifier :
 i) without filter, Ripple factor (R.F) = 1.21.
 ii) with filter, Ripple factor (R.F) = 0.052.

8/10/22



AND gate:



Truth Table:

A	B	$Y = A * B$
0	0	0
0	1	0
1	0	0
1	1	1

OR gate:

Symbol:



Truth Table:

A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate:

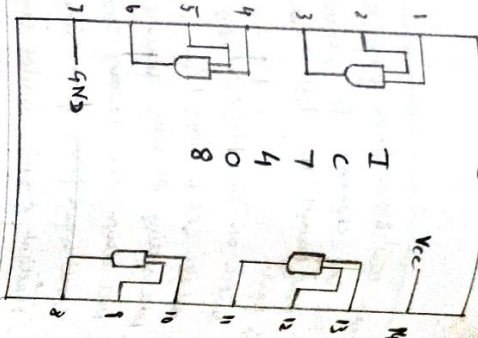
Symbol:



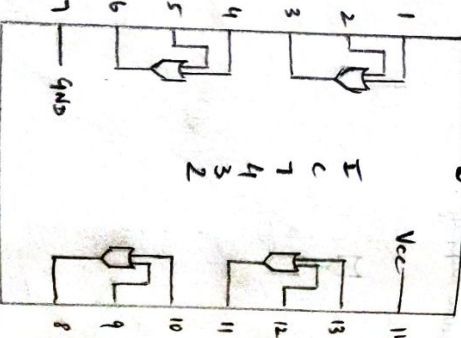
Truth Table:

A	$Y = \bar{A}$
0	1
1	0

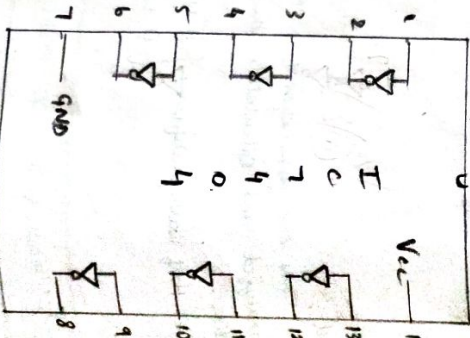
Pin diagram:



Pin diagram:



Pin diagram:



5

STUDY OF LOGIC GATES

5-8-22

Aim:

To study different logic gates and Truth table.

Components Required:

S.No	Name of the Component	No. of required
1.	IC 7400 - NAND	1 No.
2.	IC 7402 - NOR	1 No.
3.	IC 7404 - NOT	1 No.
4.	IC 7408 - AND	1 No.
5.	IC 7432 - OR	1 No.
6.	IC 7486 - XOR	1 No.
7.	Power supply	1 No.
8.	Bread Board	1 No.
9.	Connecting wires	As required.

Theory:

Logic gate is a logical decision and these are called 'logic gates'. Each gate has one or more inputs and only one output. OR, AND, NOT are basic gates. NAND, NOR and EX-OR gates are universal gates.

AND gate:

AND gate performs logical multiplication commonly known as AND function. The output is high when both inputs are high. The output is low when any one of inputs is low. IC 7408 is two input and IC 7410 is three input gates with 14 pins. Among 14 pins, 7th pin is used for GND and 14th pin is used for VCC.

OR gate:

OR gate performs logical addition known as OR function. The output is high when any of the inputs are high. The outputs are low when both inputs are low. IC 7432 is two input and one output. The logical function of OR gate is given by A and B. A consists of two gates with 14 pins and among them 7th pin is used for GND and 14th pin used for VCC.

NOT gate:

NOT gate is called inverter. The output is high when input is low. The output is low when input is high. IC 7404 is one output and one input. The logical function is given by A, A single IC 7404 consists of six gates with 14 pins, 7th pin is used for GND and 14th pin is used for VCC.



EX-OR gate:

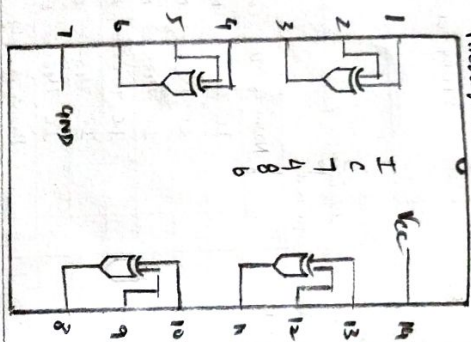
Symbol:



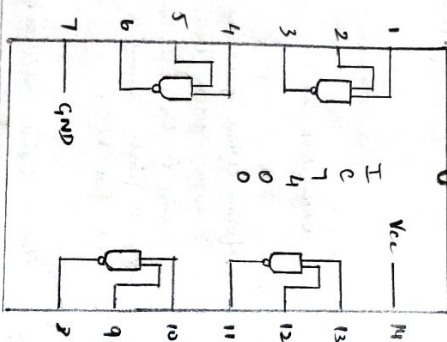
Truth Table:

A	B	$Y = A \oplus B = A \cdot \bar{B} + \bar{A} \cdot B$
0	0	0
0	1	1
1	0	1
1	1	0

Pin Diagram:



Pin Diagram:



NAND gate:

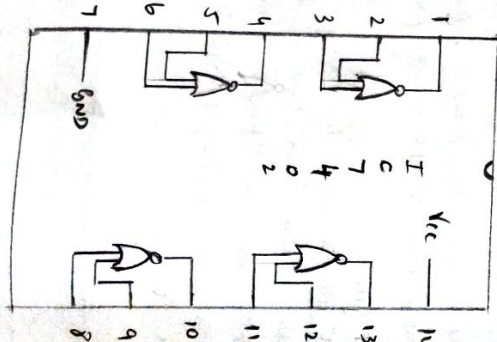
Symbol:



Truth table:

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Pin Diagram:



NOR gate:

Symbol:



Truth Table:

A	B	$Y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

NAND gate:

The NAND gate is combination of AND-NOT. The output is high when inputs are low. The output is low when inputs are high. IC7400 is two input and single IC7400 consists of four gates with 14 pins. Among them 1st pin is used for GND and 14th pin is used for VCC.

NOR gate:

NOR gate is combination of OR-NOT. The output is high when both inputs are low. The output is low when one or both inputs are high. IC7402 is two input single IC7402 consists of four gates with 14 pins. Among them 1st pin is used for GND and 14th pin is used for VCC.

EX-OR gate:

The output is high when any one of input is high. The output is low when both inputs are low and both inputs are high. IC7486 is two input and one output. The logical function of EX-OR gate is given by A and B. A single IC7486 consists of four gates with 14 pins. Among them 1st pin is used for GND and 14th pin is used for VCC.

Procedure:

- Components given as per circuit diagram
- Logical inputs are given as per circuit diagram
- observe the output and verify the truth table.

Result:

Thus the logic gates are verified using Digital Trainer kit.

[Signature]



EX-OR gate:

Symbol:



Truth Table:

A	B	$Y = A \oplus B = A \cdot \bar{B} + \bar{A} \cdot B$
0	0	0
0	1	1
1	0	1
1	1	0

NAND gate:

Symbol:



Truth table:

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

NOR gate:

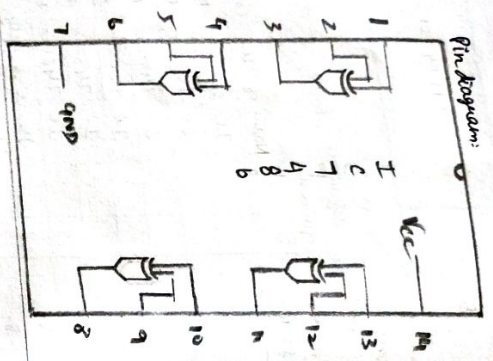
Symbol:



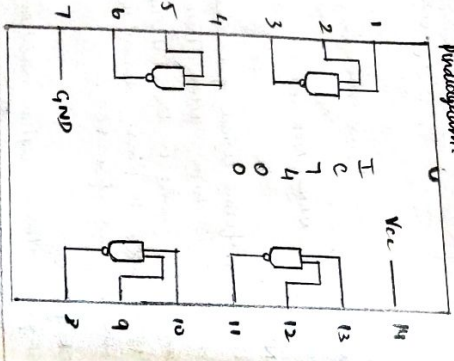
Truth Table:

A	B	$Y = \bar{A} + \bar{B}$
0	0	1
0	1	1
1	0	1
1	1	0

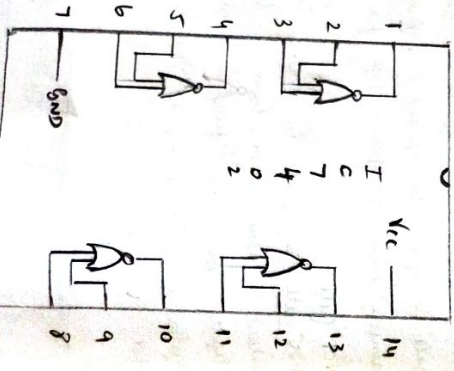
Pin diagram:



Pin diagram:



Pin diagram:



NAND gate:

The NAND gate is combination of AND-Not. The output is high when inputs are low. The output is low when inputs are high. IC 7400 is two input and one output. The logical function of NAND gate is given by $A \cdot B$. A single IC 7400 consists of four gates with 14 pins. Among them 7th pin is used for GND and 14th pin is used for VCC.

NOR gate:

NOR gate is combination of OR-Not. The output is high when both inputs are low. The output is low when one or both inputs are high. IC 7402 is two input and one output. The logical function of NOR gate is given by $\bar{A} + \bar{B}$. A single IC 7402 consists of four gates with 14 pins. Among them 7th pin is used for GND and 14th pin is used for VCC.

EXOR gate:

The output is high when any one of input is high. The output is low when both inputs are low and both inputs are high. IC 7486 is two input and one output. The logical function of EX-OR gate is given by $A \oplus B$. A single IC 7486 consists of four gates with 14 pins. Among them 7th pin is used for GND and 14th pin is used for VCC.

Procedure:

- Connections given as per circuit diagram
- Logical inputs are given as per circuit diagram
- observe the output and verify the truth table

Result:

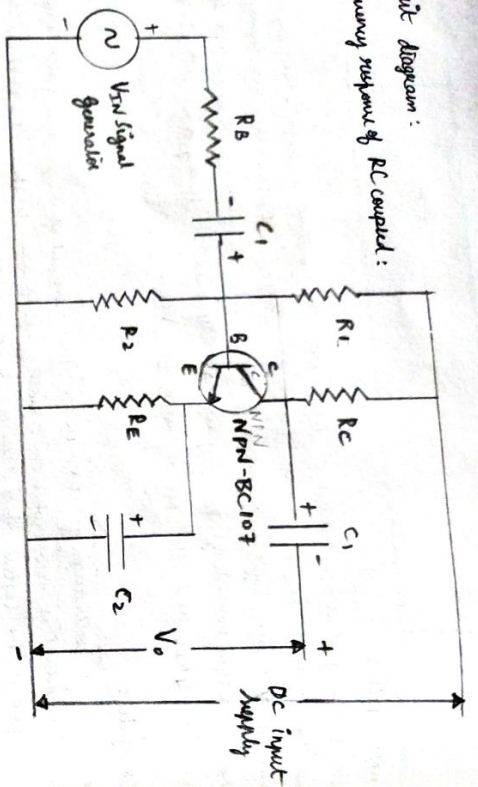
Thus the logic gates are verified using Digital Trainer kit.

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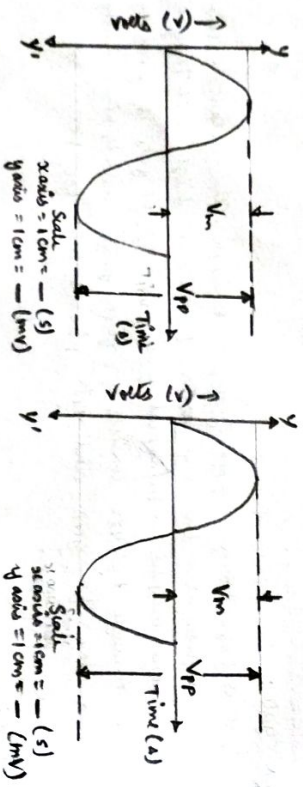
Circuit Diagram:

Frequency response of RC coupled amplifier:

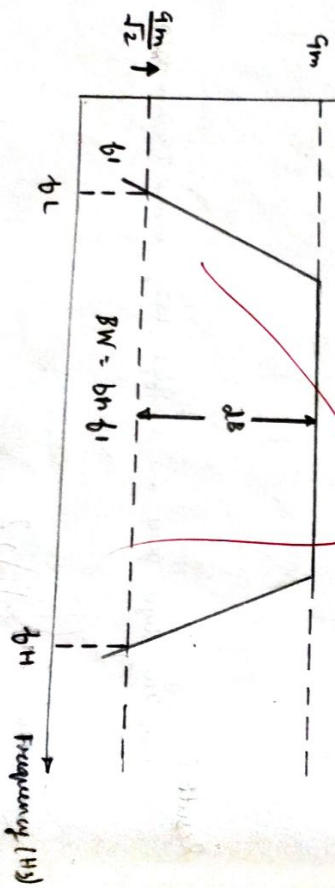


Model:

Output signal:



Frequency response of RC coupled amplifier:



STUDY OF FREQUENCY RESPONSE OF RC COUPLED AMPLIFIERS

5-8-22

Aim: To determine the frequency response of RC coupled amplifier and calculation of bandwidth.

Equipment and components required:

Sr No	Apparatus Required	Range	Quantity
1.	Transistor	BC 107	1 No.
2.	Resistor	3.3 KΩ, 2.2 KΩ, 6.8 KΩ, 10 KΩ	Each 1 No.
3.	Capacitor	4.7 μF	2 Nos.
4.	Capacitor	10 μF	1 No.
5.	Common Mode Oscillator	(0-30) MHz	1 No.
6.	Function generator	(0-5) MHz	1 No.
7.	Regulated power supply	(0-30) V	1 No.
8.	Bread board	-	1 No.
9.	Probe	-	3 Nos.
10.	Connecting wires	-	As required.

Formula Used:

$$A_v = \frac{V_{out}}{V_{in}} \quad ; \quad A_v (dB) = 20 \log \left[\frac{V_{out}}{V_{in}} \right]$$

Where V_{out} is output voltage signal, V_{in} is input signal voltage.

Precautions:

- Power switch of FG, CRO, and RPS should be kept in OFF position at time of starting.
- Connections should be neat and tight.
- Channel 1 and channel 2 of CRO should be kept at ground position at time of starting.
- Curve and fine adjustments knobs in RPS should be kept at minimum position at time of starting.

Theory:

When AC signal is applied to base of the transistor. It is amplified and developed across output of the transistor. To obtain the higher gain more wave amplifiers stages are connected in succession. It is most popular type of coupling because it provides constant amplitude at wide range of frequency where it is used for many applications like amplifier, switching devices, etc.

Frequency Response Curve:

The curve representing the variation of gain of an amplifier with frequency is known as 'Frequency Response Curve'. The frequency increases voltage gain of an amplifier. The maximum value of gain and the constant also obtain maximum value of the frequency range and afterward the gain starts decreasing and over a certain frequency range and afterward the gain starts decreasing.



with increase in frequency. It may be seen to the divide into three regions i.e. low frequency (50Hz), Mid frequency (50 to 20kHz), high frequency ($>20\text{kHz}$).

Procedure:

1. Connections are made as per circuit diagram.
2. Switch ON the CRO supply.
3. Connect the input signal to channel 1, set of CRO and select screen display mode in channel, and set the input ground position in screen display by adjusting Y-axis knob.
4. Connect the output signal to the channel 2 port and select screen display mode in channel 2 and set the output ground position in screen display by adjusting Y-axis knob.
5. Observe the waveforms, switch ON the power supply of FG and PPS.
6. Set PPS at 1V AC supply.
7. By using function generator setup the circuit at 100Hz frequency 1mV signal.
8. Set the CRO in ADD mode. Now both input and output wave with appear on the screen as shown in model graph.
9. Trace the waveform and tabulate the reading of V_{out} and V_{in} and calculate net gain (AG).
10. Tabulate the readings of various output voltage by increasing the input signal frequency from 100Hz to 2MHz .
11. Put CRO in ground position, reduce the input voltage and frequency to 100Hz and draw the frequency response graph of RC coupled amplifier between the frequency and gain.
13. Divide the curve length of the entire unit.

Result:

Thus the RC coupled amplifier is designed and implemented.



VIVA QUESTIONS

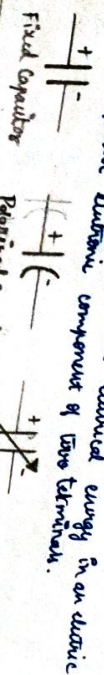
Kirchoff's laws

1. What is electric current?
The flow of electrons is called electric current [unit - ampere (A)].
2. Define Electric Potential.
The capacity of charged body to work is called electric potential (unit - Volt).
3. What is resistance?
The property of material by virtue that opposes or resists the flow of current is called resistance [unit - ohm (Ω)].
4. Define electrical energy.
The rate of work done in electrical circuit is called electrical energy. It is denoted by 'E'.
5. What is electrical power?
The rate of work done in electrical circuit is called electrical power. It is denoted by 'P' [unit - Watt].
6. On what factor the resistance offered by conductor depends on?
- It is directly proportional to length of conductor
- It is inversely proportional to the cross-section of conductor
- Dependent on temperature of conductor
- Dependent on nature of material of conductor
7. What is conductance (G) or its law?
It states that, "For any conductor potential difference (V) between two ends and current (I) flowing between two ends of conductor are directly proportional to each other. $V = IR$ (or) $\frac{V}{R} = I$ (or) $R = \frac{V}{I}$.
8. What is conductance?
The property of conductor due to which it allows the current through pass through its known as conductance. It is represented as 'G'.
9. What Kirchoff's Voltage law?
The algebraic sum of product of current and resistance is equal to electromotive force in the circuit.
In other words, the IR drop across the circuit is equal to voltage drop (induced emf).
$$\sum E = \sum V_1 + V_2 \quad (or) \quad E = V_1 + V_2$$
10. What Kirchoff's Current law?
The algebraic sum of incoming current is equal to sum of outgoing current.
$$\sum I_1 - \sum I_2 - \sum I_3 = 0 \quad (or) \quad I_1 = I_2 + I_3$$



12) What is Inductor?

An Inductor is also called an coil, store a magnetic field when electric current component that store energy in a magnetic field. It is a passive electronic component of two terminals.



PN Junction diode.

1. What is semiconductor?

Substance with conductance property between metals and insulators. Two types are Intrinsic semiconductor and Extrinsic semiconductor.

2. What is Intrinsic semiconductor?

A pure semiconductor completely free from impurities is called intrinsic semiconductor.

3. Extrinsic semiconductor?

A semiconductor doped by group 13 and group 15 has is called extrinsic semiconductor.

4. What is meant by doped semiconductor?

Extrinsic semiconductor is called as doped semiconductor.

5. What is doping?

The process of addition of impurities to pure substance in order to alter its conduction properties is called doping.

6. What are two types of impurities?

The impurities are of P-Type and N-Type, two types respectively.

7. Effect of temperature on conductivity of semiconductor?

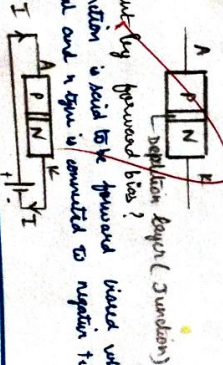
It increases while increase in temperature.

8. What is Junction diode?

Junction diode is formed when P-type and N-type semiconductors is in intimate contact with each other.

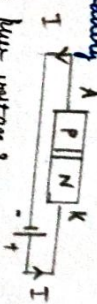
9. What is meant by forward bias?

A PN Junction is said to be forward biased when P-region is connected to positive terminal and n-type is connected to negative terminal of battery.



10. What is meant by Reverse Bias?

A PN Junction diode is said to be reverse biased when P-region is connected to negative terminal and n-region is connected to positive terminal of the battery.



11. What is knee voltage?

In forward bias, the voltage across the PN-junction is very low and current is small. When the voltage is increased, the current increases sharply. This voltage is called knee voltage.

12. What is meant by Reverse Breakdown?

For a certain value of reverse voltage, the PN-junction diode starts conducting. This is called reverse breakdown.

13. Application of diode?

- LEDs are used in lamps and laser diodes
- Zener diodes are used as voltage regulators
- Diodes are used in rectifying AC to DC
- They are used in switching signals, i.e., switching on and off.

Application of PN Junction diode as half-wave rectifier.

1. What is Rectifier?

A rectifier is an electrical device that converts alternating current (AC) into direct current (DC).

2. What are the types of Rectifiers?

- Half wave rectifier
- Full wave rectifier

3. What is half wave rectifier?

A half wave rectifier converts an AC signal to DC signal by allowing only the positive half cycle of the waveform to pass and blocking the negative half cycle.

4. What is full wave rectifier?

A full wave rectifier is a circuit that converts both the positive and negative half cycles of an AC signal into a single polarity (DC) output.

5. What is form factor?

The form factor is the ratio of RMS value to the average value of the waveform.

For a half wave rectifier, the form factor is 1.57.

6. What is ripple factor?

Ripple factor = $\frac{\text{RMS value of AC component}}{\text{DC value of DC component}}$

For half wave rectifier, the ripple factor is 1.21.



7. What is peak factor?

The peak factor is defined as ratio of maximum value to RMS value of an alternating quantity.

$$V_{rms} = \frac{V_m}{2}$$

8. Applications of Rectifier!

- DC welder
- DC motor drive
- Battery charger
- DC power supply
- HVDC.

9. What is efficiency of Rectifier?

The efficiency of rectifier is ratio of output DC power to input AC power. For a half wave rectifier the efficiency is 40.6%.

10. What is meant by filter in Rectifier?

The filter is a device that allows the passing of AC component of rectifier output. The filter circuit can be constructed by combination of components like capacitor, inductor and Inductors.

RC coupled Amplifier

1. Define amplifier? What is tripling?

An amplifier is a electronic device that increases the voltage current or power of a signal. Eg. wireless communication and broadcasting audio components.

2. What is coupled amplifier?

Amplifier in which two or more stages connected together. In the output of one stage forms as input to another stage. It called coupled amplifier.

3. How many types of coupled amplifiers you know?

- RC coupled amplifier

- Transformer coupled amplifier

- LC coupled amplifier.

4. Why it is RC coupled amplifier?

This is because the resistance R_c and coupling capacitor C_c are responsible for carrying output of 1st stage to input of next stage.

5. Define voltage gain and power gain of amplifier?

$$\text{Voltage gain (A}_v\text{)} = \frac{\text{Output voltage (V}_{out}\text{)}}{\text{Input voltage (V}_{in}\text{)}}.$$

10/8/22





Completed
8/6/8/22.

